

CUPRIJA, Serbia

WMO: 133840

Lat: 43.933N

Lon: 21.383E

Elev: 410

StdP: 14.48

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.3	17.3	6.9	7.9	14.2	11.8	10.2	19.1	16.8	32.4	14.3	31.9	2.1	320	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.6	95.2	70.6	91.6	69.5	88.2	68.7	73.6	88.4	71.9	86.4	70.5	84.1	4.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.2	109.1	78.9	67.5	102.5	77.1	66.0	97.3	75.6	37.3	88.2	35.8	86.3	34.6	84.2	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.1	10.8	9.0	3.6	100.6	7.6	4.0	-1.8	103.5	-6.3	105.9	-10.5	108.1	-16.1	111.0		
(5)			3.0	76.2	7.4	1.8	-2.3	77.5	-6.6	78.6	-10.8	79.6	-16.2	80.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	33.2	36.8	44.6	54.0	62.2	68.7	72.2	72.2	63.6	53.7	45.0	35.6		
	DBStd	15.66	8.59	9.20	8.37	7.27	6.42	6.33	5.56	5.77	6.91	7.90	8.30	8.24		
	HDD50	1849	522	376	209	42	2	0	0	0	55	193	450			
	HDD65	4900	986	791	634	334	130	35	7	10	107	355	600	911		
	CDD50	3154	1	6	40	163	380	560	689	687	408	171	44	5		
	CDD65	726	0	0	0	5	43	145	231	232	64	6	0	0		
	CDH74	8730	0	0	7	108	604	1716	2759	2701	753	81	1	0		
	CDH80	3697	0	0	0	22	153	680	1282	1293	256	11	0	0		
	WSAvg	3.4	3.6	3.7	3.9	3.8	3.2	3.0	3.0	2.8	2.9	3.1	3.8	3.6		
	PrecAvg	26.7	1.9	1.8	1.9	2.5	2.8	3.0	2.4	1.8	2.3	2.2	2.0	2.2		
(15) Precipitation	PrecMax	39.2	4.5	3.8	4.4	5.8	7.1	6.4	7.6	5.9	5.4	4.7	4.8	4.3		
	PrecMin	17.5	0.5	0.6	0.2	0.2	0.7	0.8	0.5	0.0	0.4	0.1	0.0	0.3		
	PrecStd	5.4	1.1	0.8	1.1	1.2	1.5	1.5	1.5	1.3	1.3	1.3	1.1	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	66.5	74.2	83.1	88.2	95.1	99.7	100.7	93.3	81.1	70.3	60.6		
		MCWB	48.8	52.0	55.3	60.9	66.4	71.8	71.6	71.1	67.7	63.9	57.9	52.1		
	2%	DB	52.1	60.0	69.1	76.6	84.0	90.8	95.1	95.7	86.4	75.8	66.0	55.3		
		MCWB	46.2	49.5	53.1	59.0	65.2	70.5	70.3	70.4	66.0	61.9	56.0	49.2		
	5%	DB	48.5	55.2	64.1	72.4	80.8	87.5	91.3	91.5	82.1	71.7	61.7	50.6		
		MCWB	44.1	46.8	50.8	56.7	64.0	69.4	69.8	69.2	64.6	60.1	54.0	46.1		
	10%	DB	45.0	50.7	59.4	68.3	77.0	83.8	87.4	87.5	77.8	67.5	57.5	46.7		
		MCWB	41.8	44.8	49.0	55.7	62.7	68.2	69.1	68.3	63.2	58.2	51.6	43.4		
	0.4%	WB	49.4	53.7	57.3	63.6	69.4	75.5	75.3	75.6	70.0	66.2	59.3	53.5		
		MCDB	55.2	63.7	70.2	77.9	82.5	88.4	90.6	91.3	86.5	77.0	67.5	58.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.8	50.3	54.1	60.7	67.0	72.9	73.3	73.1	67.5	63.0	56.8	49.5		
		MCDB	51.7	58.2	65.5	73.9	79.7	86.4	88.5	88.1	81.6	72.8	64.4	54.7		
	5%	WB	44.4	47.6	52.0	58.3	65.5	71.0	71.8	71.3	65.9	61.0	54.6	46.4		
		MCDB	48.1	54.3	62.3	69.4	77.6	83.5	86.8	86.2	77.9	69.9	60.9	50.5		
	10%	WB	41.9	45.0	49.8	56.4	63.8	69.3	70.3	69.7	64.4	58.9	52.0	43.8		
		MCDB	44.8	50.4	58.5	66.1	74.6	81.4	83.9	83.4	75.1	66.3	57.1	46.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	15.8	20.0	21.9	22.8	23.8	25.6	26.6	22.7	19.8	15.3	11.8		
		MCDBR	19.0	25.9	30.1	30.8	30.4	31.0	33.6	34.7	30.6	27.1	21.9	18.0		
		MCWBR	13.4	15.9	15.8	14.9	13.3	12.5	12.0	11.9	12.3	14.1	12.8	12.4		
		MCDBR	17.8	23.2	26.0	26.4	26.6	27.3	28.3	28.8	26.2	24.7	19.9	16.6		
	5% WB	MCWBR	12.9	14.8	14.6	13.6	12.7	12.0	11.8	11.4	11.8	13.7	11.9	11.9		
(40) Clear-Sky Solar Irradiance	taub		0.345	0.373	0.411	0.471	0.459	0.451	0.453	0.467	0.444	0.402	0.365	0.342		
	taud		2.301	2.205	2.174	2.030	2.126	2.194	2.190	2.148	2.178	2.284	2.335	2.330		
	Ebn at Noon		240	253	259	252	260	262	259	251	246	240	231	229		
	Edh at Noon		27	35	41	51	48	45	45	45	41	32	26	24		
(44) All-Sky Solar Radiation	RadAvg		417	680	1053	1420	1720	1940	1985	1781	1247	850	527	362		
	RadStd		61	91	136	164	177	189	154	170	122	114	59	53		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.22	N/A	N/A	N/A	+1.12	+1.30	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)		+1.19	N/A	N/A	N/A	+0.72	+0.76	N/A	-275	+182	N/A

Nomenclature: See separate page